

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Eight Semester of B. Tech. (EE) Examination

May 2014

EE-415 Advances in Power System

Date: 12.05.2014, Monday

Time: 01:30 p.m. To 04:30 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I

- Q - 1 (a) Define the term Phasor Measurement Unit. Draw and explain basic block diagram of PMU. [07]
- Q - 2 (a) Obtain the expressions for average direct voltage, Peak-to-peak ripple, PIV and transformer current on the secondary side in a 6-pulse Graetz's converter circuit with delay angle α . [07]
- (b) Give the difference between HVDC link and EHVAC link. [07]

OR

- Q - 2 (a) Explain power flow in HVDC links. Also derive the expression of power loss in DC system and power in the middle of the DC line. [07]
- (b) With a neat schematic diagram, state the various apparatus required for HVDC station and explain the purpose of each. [07]
- Q - 3 (a) Write short notes on multi-bridge converter with necessary diagram. [07]
- (b) A bipolar two terminal HVDC link is delivering 1000 MW at ± 500 KV at The receiving end. The total losses in the D.C. circuit are 50 MW. Calculate the following: [07]
- (i) Sending end power
 - (ii) Sending end voltage
 - (iii) Power in the middle of the line
 - (iv) Voltage in the middle of the line
 - (v) Total resistance of the D.C. circuit.

OR

- Q - 3 (a) Draw and explain ideal converter control steady-state characteristics of HVDC system. [07]
- (b) An HVDC bipolar link using a six pulse converter is supplying power of 500 MW at [07]

± 100 KV. The rectifier station is working at a delay angle $\alpha = 15^\circ$. Estimate the following:-

- (i) No load DC voltage,
- (ii) AC voltage on the converter side of the transformer
- (iii) Transformer secondary rating
- (iv) Peak inverse voltage
- (v) Peak-to-Peak voltage
- (vi) Current on the secondary side of the converter transformer.

SECTION - II

Q - 4 (a) Give comparison between series compensation and shunt compensation. [04]

(b) Choose the correct alternative: [03]

(1) In Mono-polar system the power rating is almost _____ the rating of Bi-polar system.

(a) Double (b) Half (c) 4 times (d) Equal

(2) Inverter mode in HVDC system is operated with _____ delay angle.

(a) Greater than 90° (b) Less than 90° (c) Zero (d) 30°

(3) A _____ HVDC transmission system has two poles of same polarity and return earth.

(a) Mono-polar (b) Homo-polar (c) Bi-polar (d) Multi-terminal

Q - 5 (a) Draw and explain operating characteristics of a FC-TCR with and without step down transformer. [07]

(b) What are the different modes of TCSC operations? Explain each of them in detail. [07]

OR

Q - 5 (a) Compare characteristics of SVC and STATCOM with necessary diagrams and usual notations. [07]

(b) Sketch schematic diagram and explain characteristics of UPFC in power system. [07]

Q - 6 (a) What is fuel cell? State characteristics of fuel cell. [06]

(b) What is smart grid? Explain benefits and features of smart grid. [08]

OR

Q - 6 (a) What is wrong with existing grid? Explain elements of smart grid. [08]

(b) Explain advantages and disadvantages of fuel cell according to their types. [06]
